Project Report : **HADOOP_SENTIMENT_ANALYSIS**

DataSet Name: Twitter Reddit social media dataset (kaggle)

DataSet Size: 1.6 Millions (roughly 200-300 MB)

[Two ways to perform this Analysis:

1) Jupyter Notebook (faster and easier)

2) Hadoop MapReduce(Java/Python) via HDFS is fully supported on Linux Ubuntu

Which Method I followed?

➔ 2)

What System I used? (Requirements)

➔ My Laptop (Ram: 16 GB, 64 bit OS)
➔ WSL Linux/Ubuntu 20.x.x setup

Pre_set-up: [** note: SKIP IF ALREADY DONE **]

Pre-requesites for Hadoop:

➔ Install Networking Tools ($ sudo apt install net-tools)
➔ Install Gedit ($ sudo apt gedit)
➔ Install Open SSH ($sudo apt-get install openssh-server && openssh-client)
➔ Install JAVA ($ sudo apt install openjdk-8-jdk -y)
➔ Install MySQL ($ sudo apt install mysql-server -y)
➔ Disable IPV6
➔ Hadoop version – 3.3.6 (Download the tar file and untar it in a dedicated user)
➔ Configure the Hadoop files (~/hadoop/etc/hadoop/nano (filename))
➔ Python version (3.11.x)

What will Approach 2) will do?

➔ The Mapper: Reads the CSV line by line, cleans the tweet/post text, and uses a lightweight lexicon to calculate a score. It outputs (sentiment_category, 1).
➔ The Reducer: Collects all the keys and sums them up to output the final count: (positive, 54300), (negative, 32100).

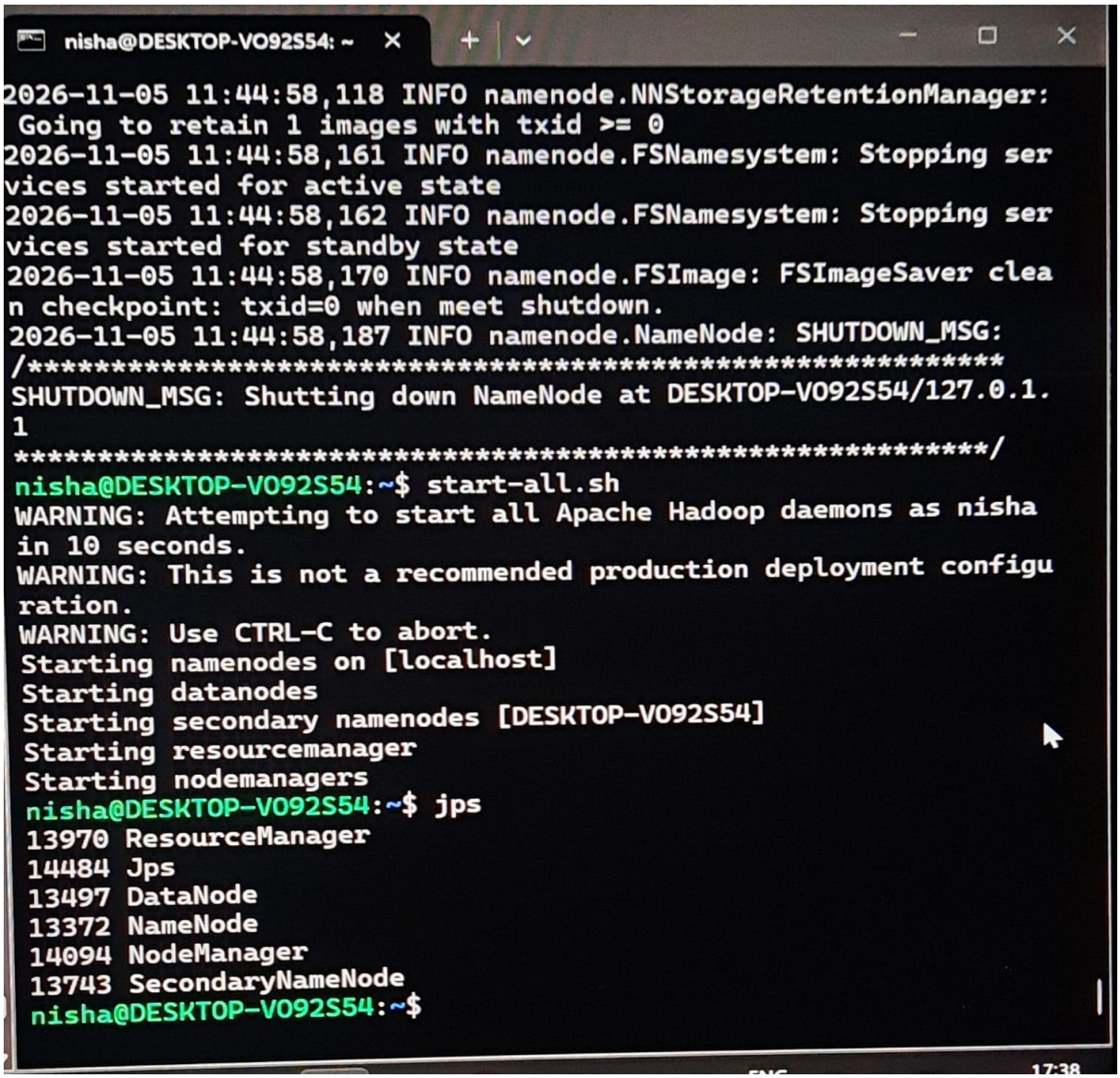
Why This Approach: To showcase Hadoop/Big Data infrastructure skills on your resume, or if the dataset is too massive to fit into the computer's RAM.

Task 1: Data Acquisition

First of all start Hadoop services:

$ start-all.sh

Run jps



All the required node has started successfully!!

(** NOTE- I have done this project using pseudo node i.e. the master and the slave remains on the same system providing you the same compatible environment)

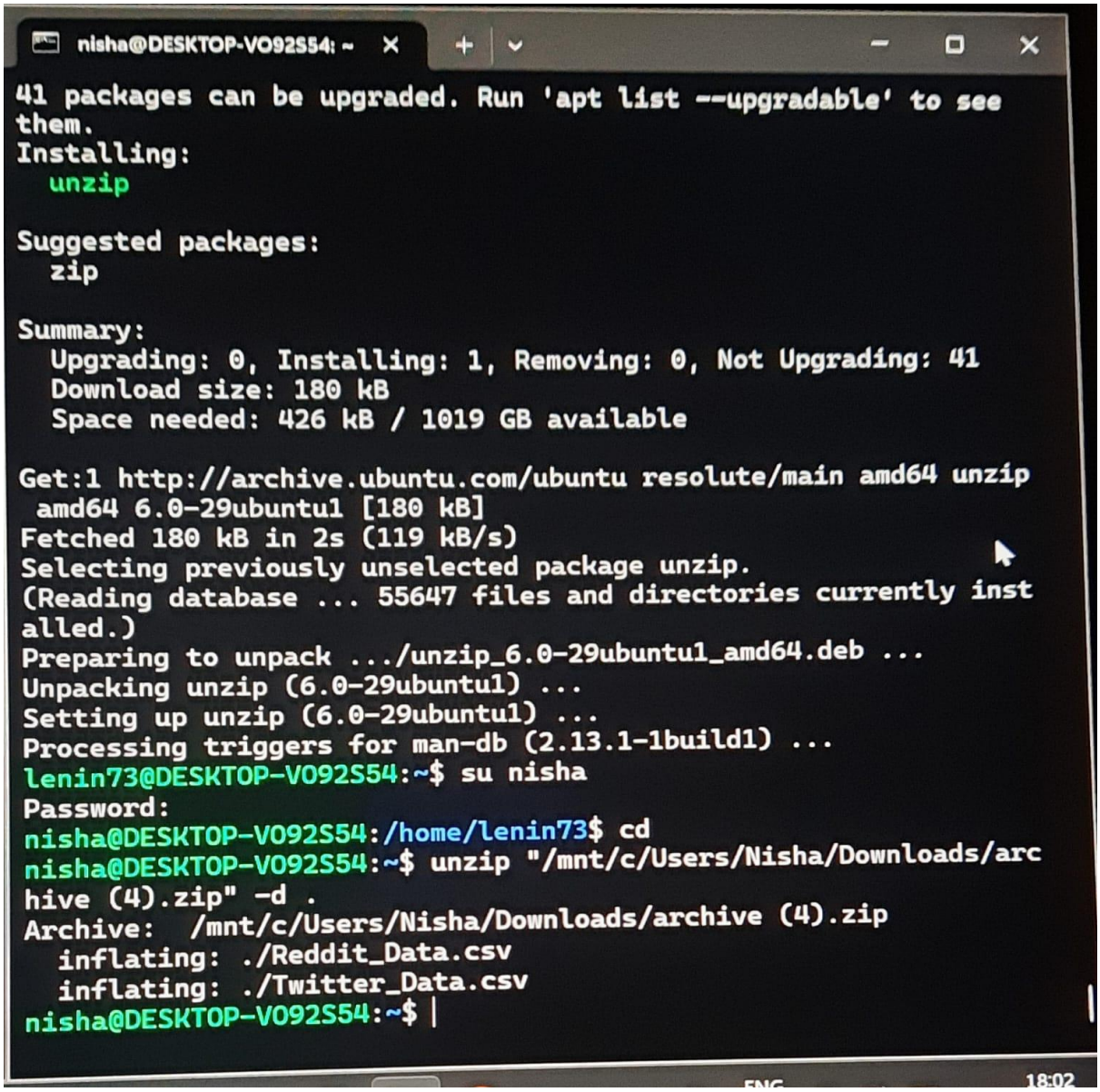
Step2.a) Make a directory (~/sentiment_project)

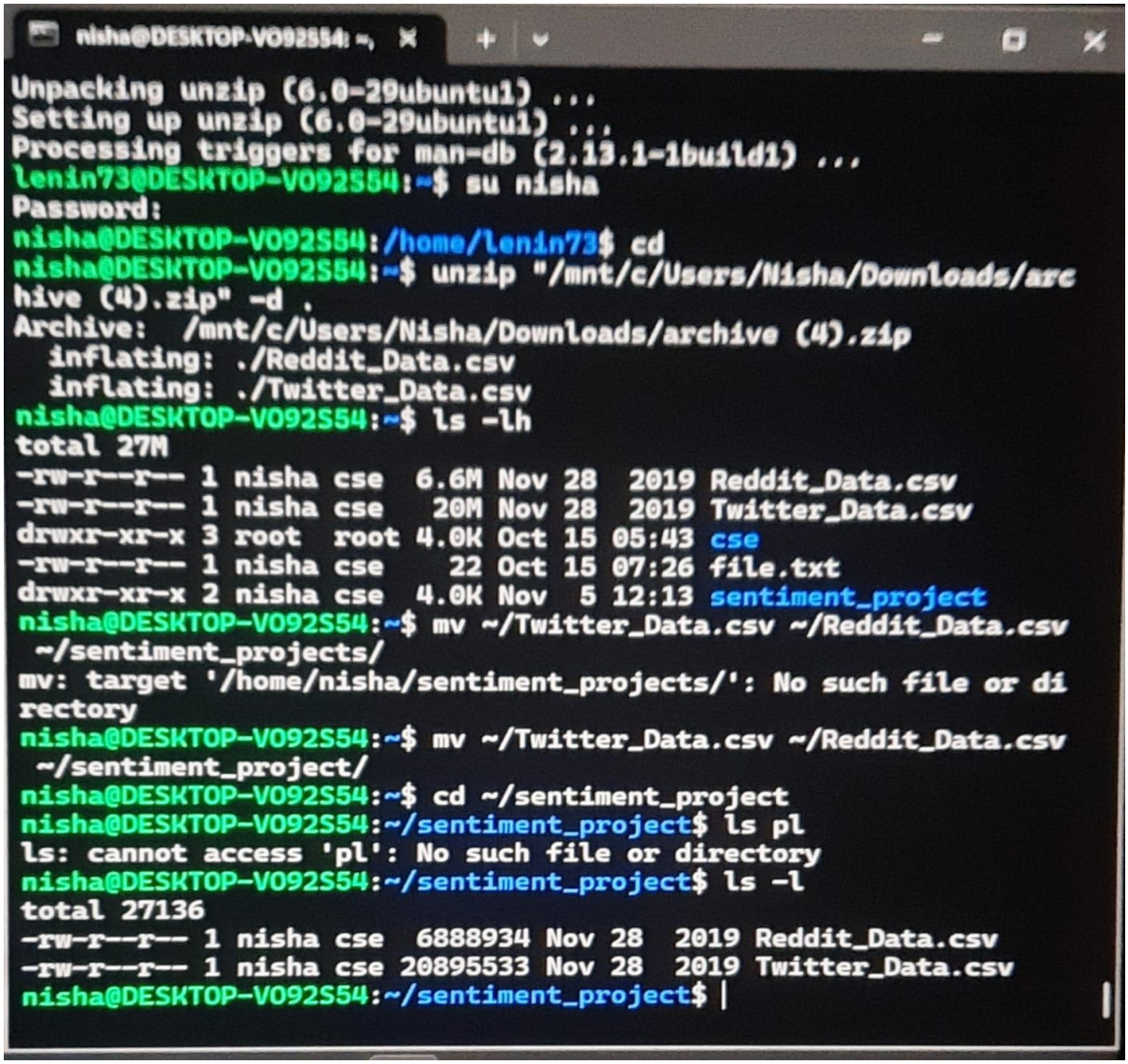
$ sudo mkdir -p ~/sentiment_project

Go to the directory: cd ~/sentiment_project

2.b) unzip the zipped dataset file

$ unzip /mnt/c/Users/$(cmd.exe /c "echo %USERNAME%" 2>/dev/null | tr -d '\r')/Downloads/*.zip -d ~/sentiment_project/


nisha@DESKTOP-VO92S54: ~
41 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
unzip
Suggested packages:
zip
Summary:
Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 41
Download size: 180 kB
Space needed: 426 kB / 1019 GB available
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 unzip amd64 6.0-29ubuntu1 [180 kB]
Fetched 180 kB in 2s (119 kB/s)
Selecting previously unselected package unzip.
(Reading database ... 55647 files and directories currently installed.)
Preparing to unpack .../unzip_6.0-29ubuntu1_amd64.deb ...
Unpacking unzip (6.0-29ubuntu1) ...
Setting up unzip (6.0-29ubuntu1) ...
Processing triggers for man-db (2.13.1-1build1) ...
lenin73@DESKTOP-VO92S54:~$ su nisha
Password:
nisha@DESKTOP-VO92S54:/home/lenin73$ cd
nisha@DESKTOP-VO92S54:~$ unzip "/mnt/c/Users/Nisha/Downloads/archive (4).zip" -d .
Archive: /mnt/c/Users/Nisha/Downloads/archive (4).zip
inflating: ./Reddit_Data.csv
inflating: ./Twitter_Data.csv
nisha@DESKTOP-VO92S54:~$
ENG
18:02



Step3) Create the input directory in HDFS:

$ hdfs dfs -mkdir -p /sentiment/input

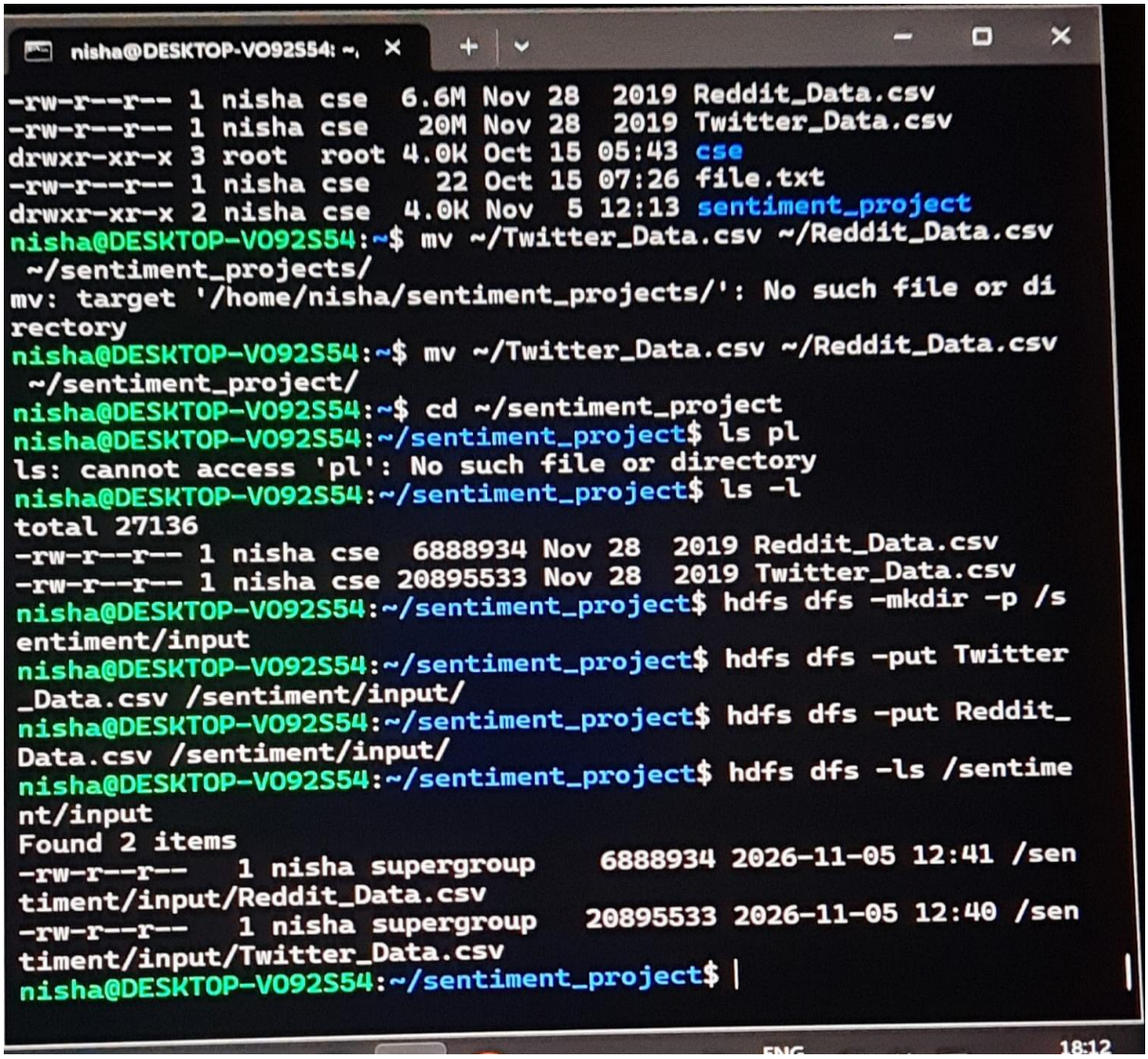
Next, upload the csv file to HDFS

hdfs dfs -put ~/sentiment_project/Twitter_Data.csv /sentiment/input/

hdfs dfs -put ~/sentiment_project/Reddit_Data.csv /sentiment/input/

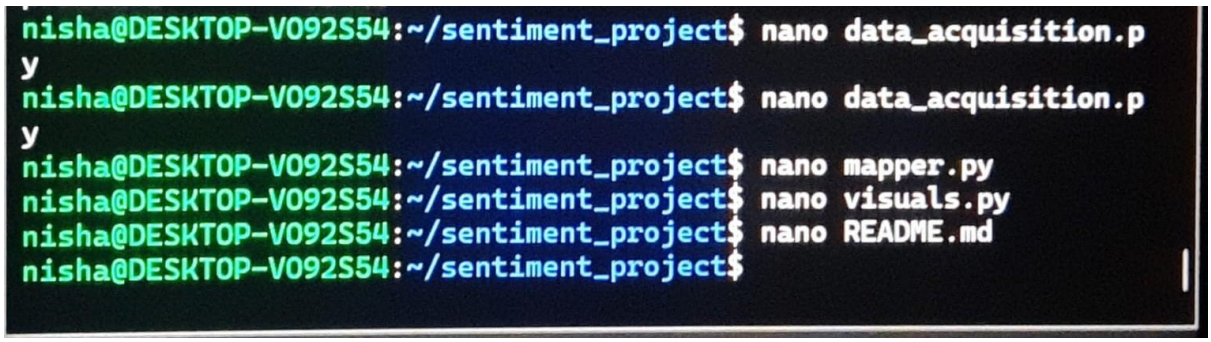
now, Verify the Data if it’s present in HDFS or not?

hdfs dfs -ls /sentiment/input



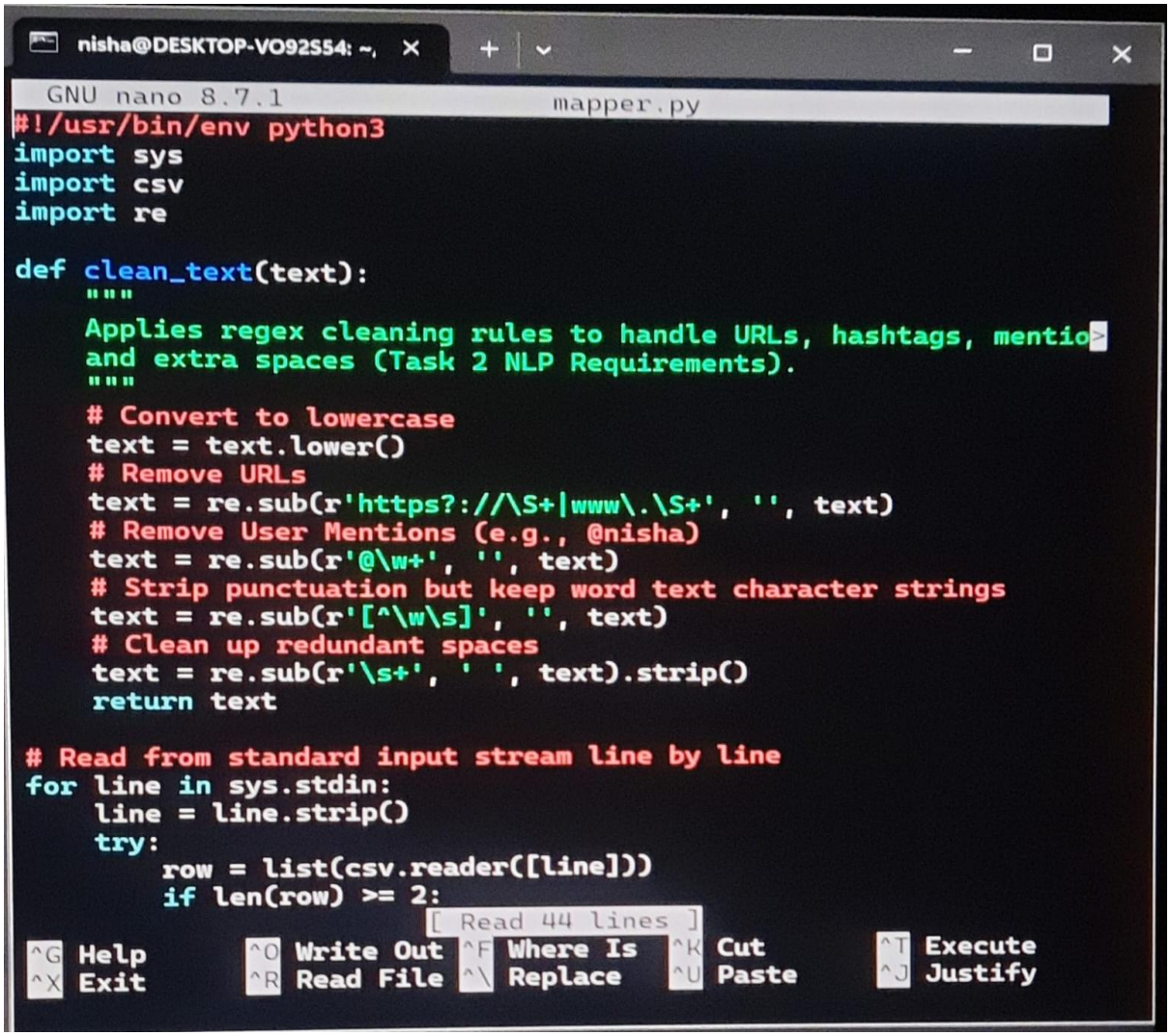
For Data acquisition create data_acquisition.py file:

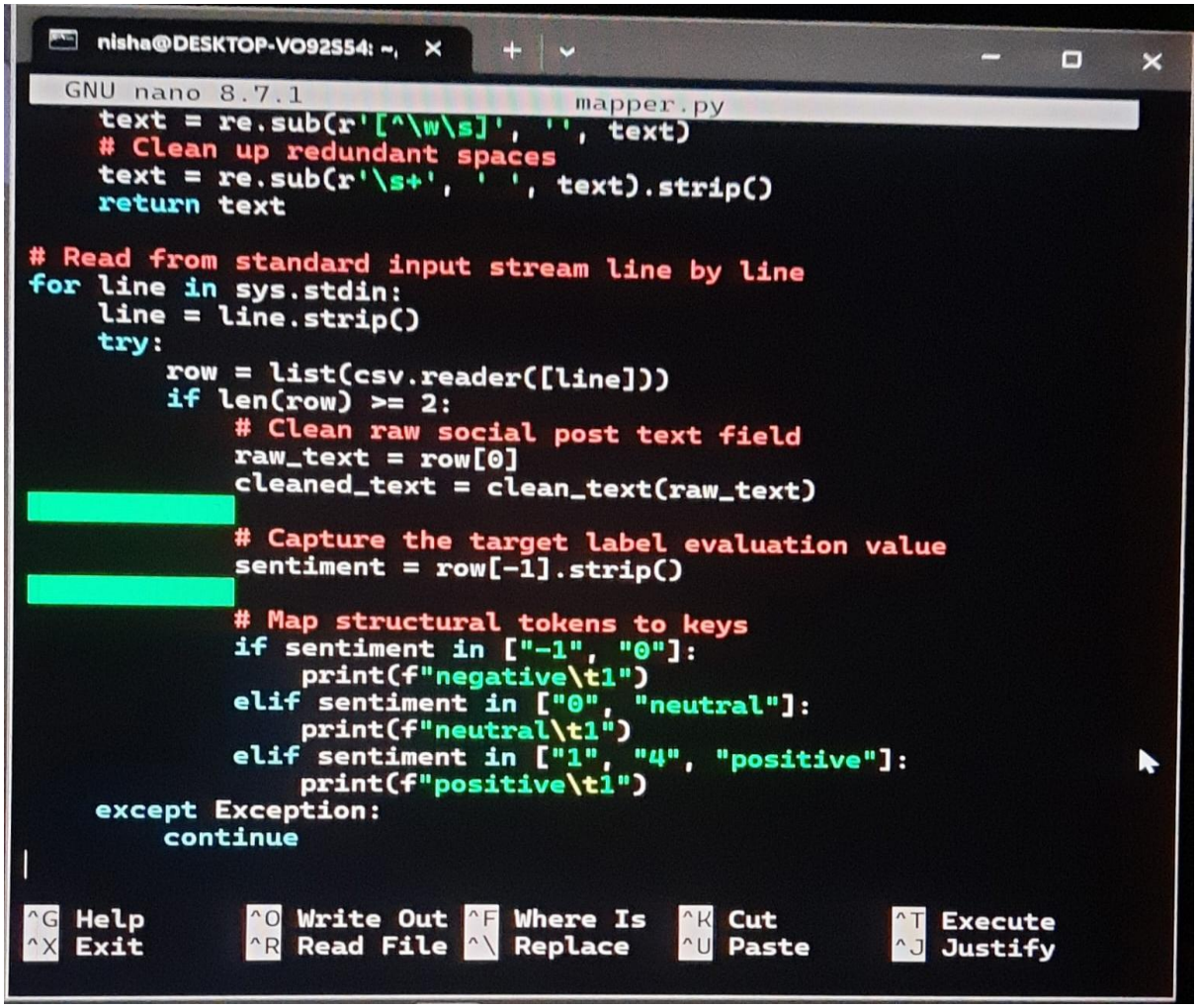
Nano data_acuisition.py



Now Finally , Create the Mapper (mapper.py)

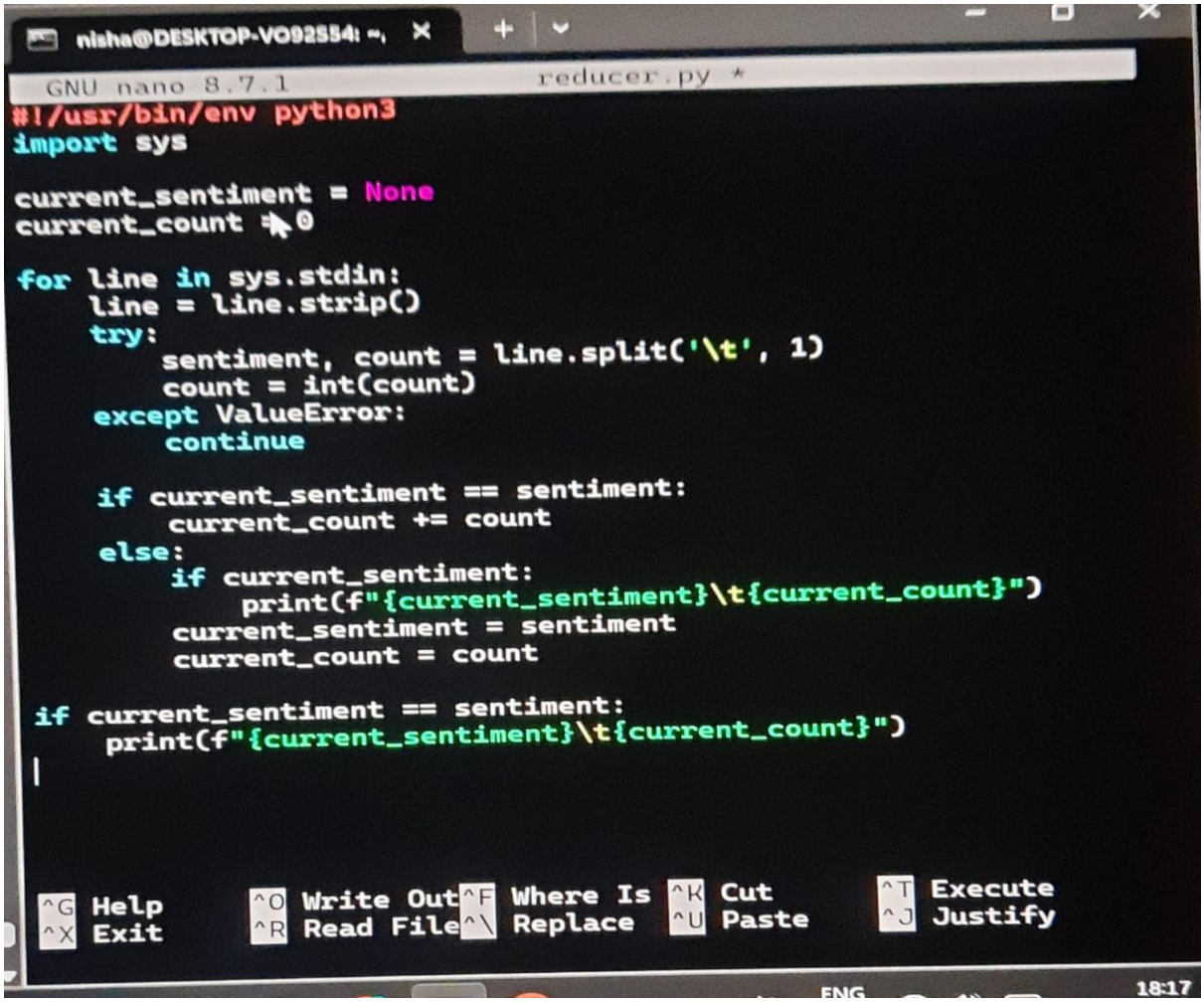
$ nano mapper.py


nisha@DESKTOP-VO92S54: ~
GNU nano 8.7.1
mapper.py
#!/usr/bin/env python3
import sys
import csv
import re
def clean_text(text):
"""
Applies regex cleaning rules to handle URLs, hashtags, mentio
and extra spaces (Task 2 NLP Requirements).
"""
Convert to lowercase
text = text.lower()
Remove URLs
text = re.sub(r'https?://\S+|www\.\S+', '', text)
Remove User Mentions (e.g., @nisha)
text = re.sub(r'@\w+', '', text)
Strip punctuation but keep word text character strings
text = re.sub(r'[^\w\s]', '', text)
Clean up redundant spaces
text = re.sub(r'\s+', ' ', text).strip()
return text
Read from standard input stream line by line
for line in sys.stdin:
line = line.strip()
try:
row = list(csv.reader([line]))
if len(row) >= 2:
[Read 44 lines]
^G Help
^O Write Out
^F Where Is
^K Cut
^T Execute
^X Exit
^R Read File
^\ Replace
^U Paste
^J Justify



Next , create the Reducer (reducer.py)

Nano reducer.py

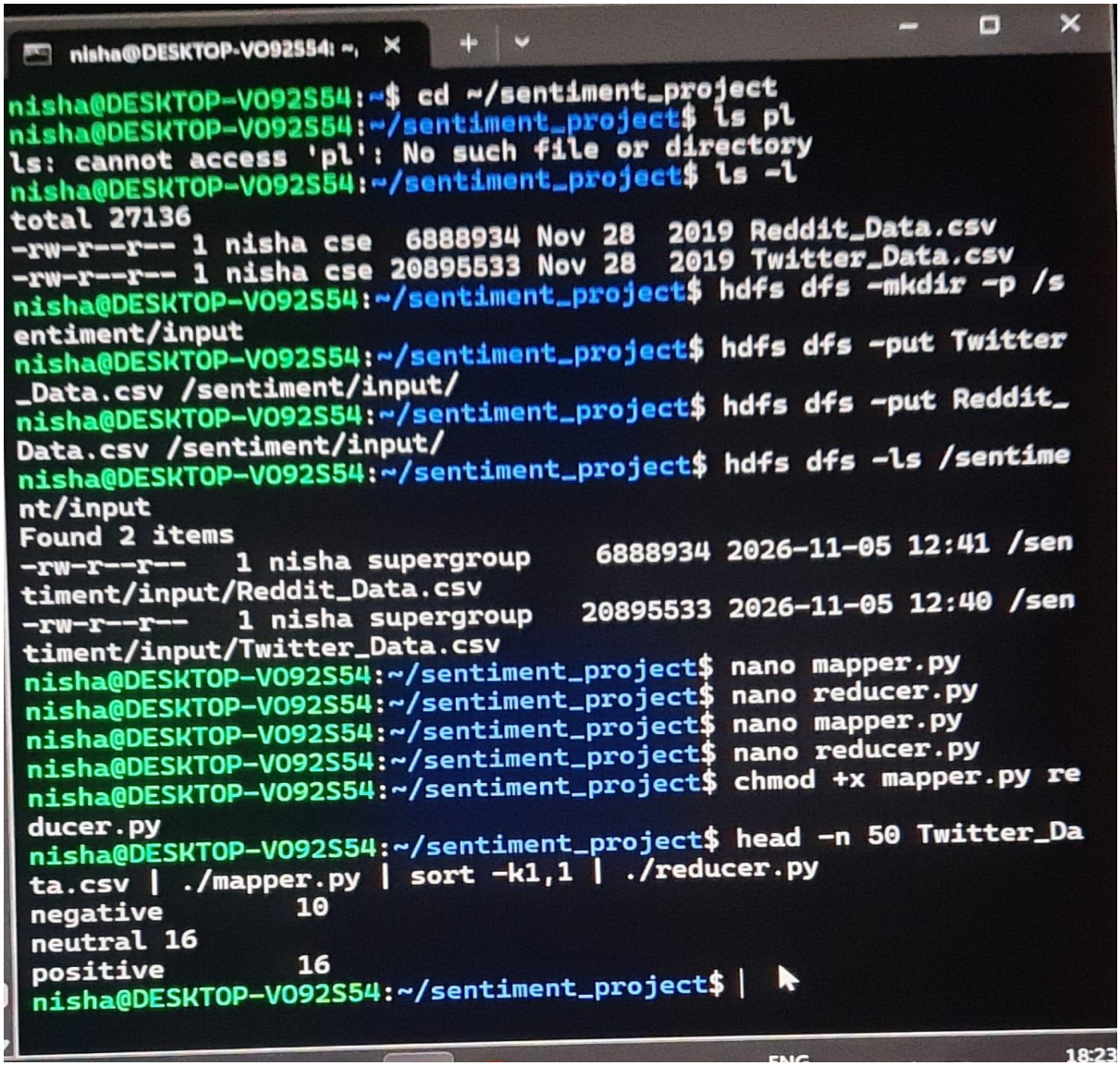


At first let's check it by running locally (for safety, so that no huge error or bug have any chance)

chmod +x mapper.py reducer.py

(Local dry run)

head -n 50 Twitter_Data.csv | ./mapper.py | sort -k1,1 | ./reducer.py



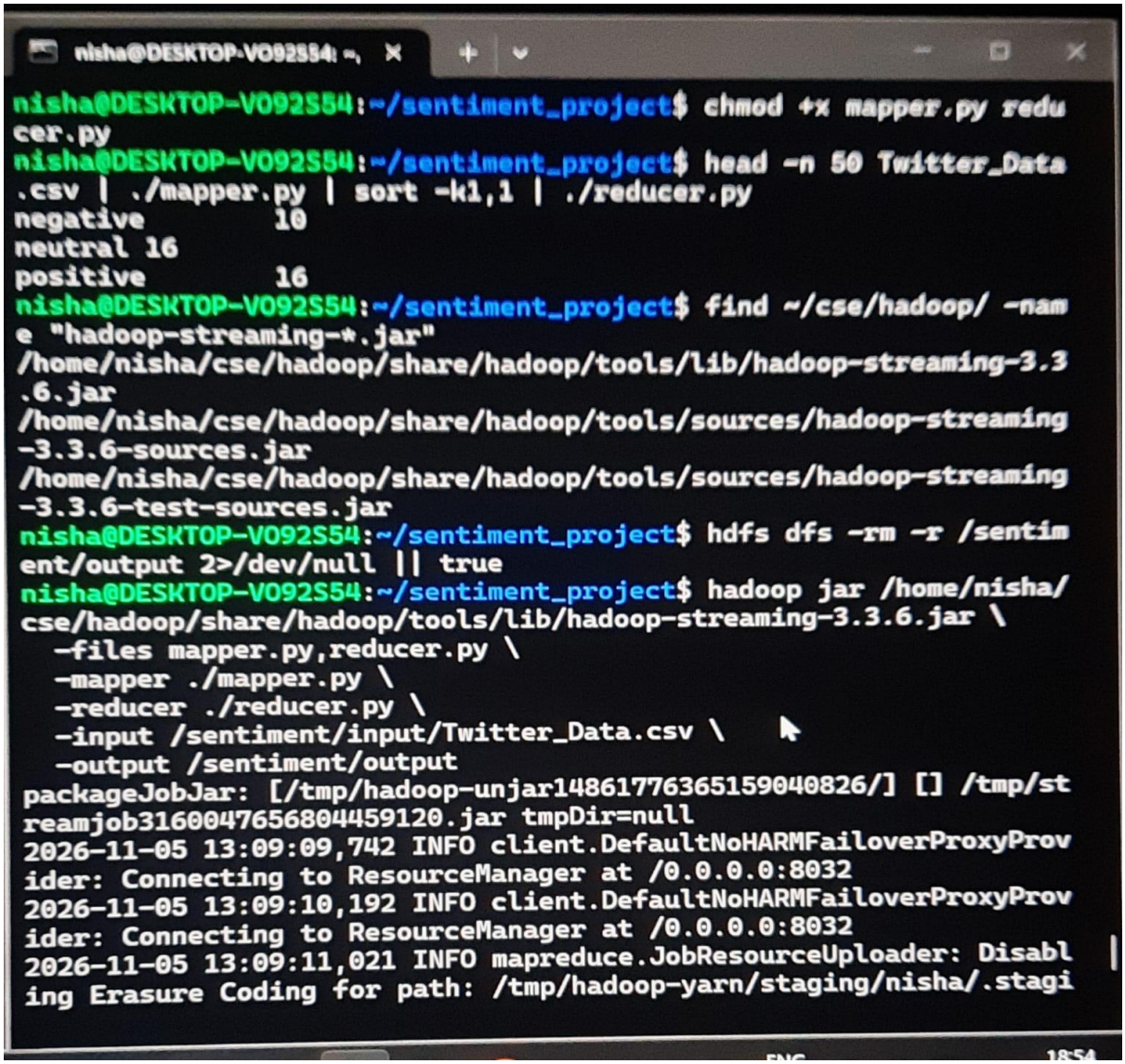
As locally it runned flawlessly, let's stream it to Hadoop (jar file):

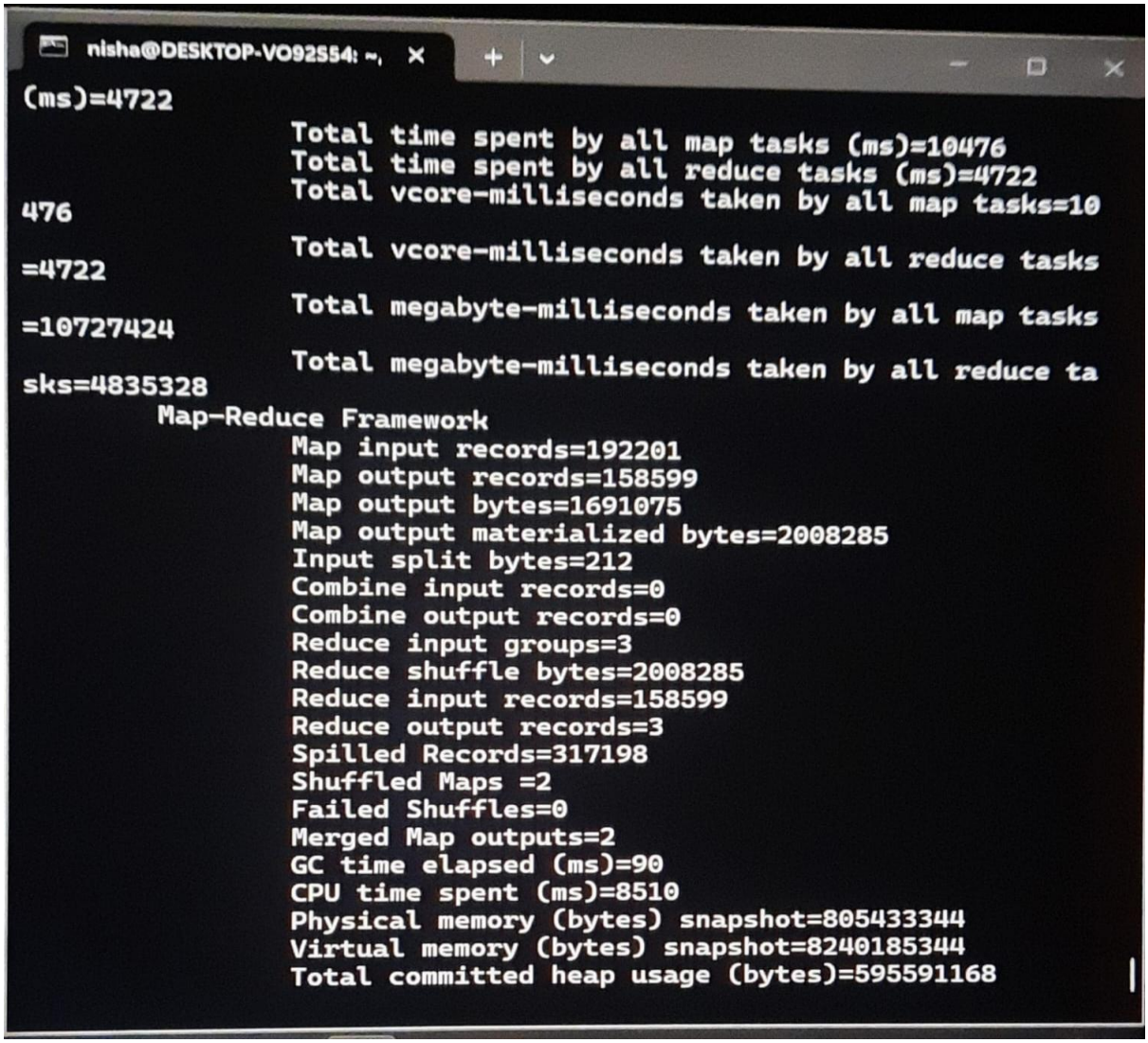
Task 2: (Text Preprocessing and NLP preparation) [jupyter notebook]

Task 3: (Sentiment Analysis Modeling) [jupyter Notebook]

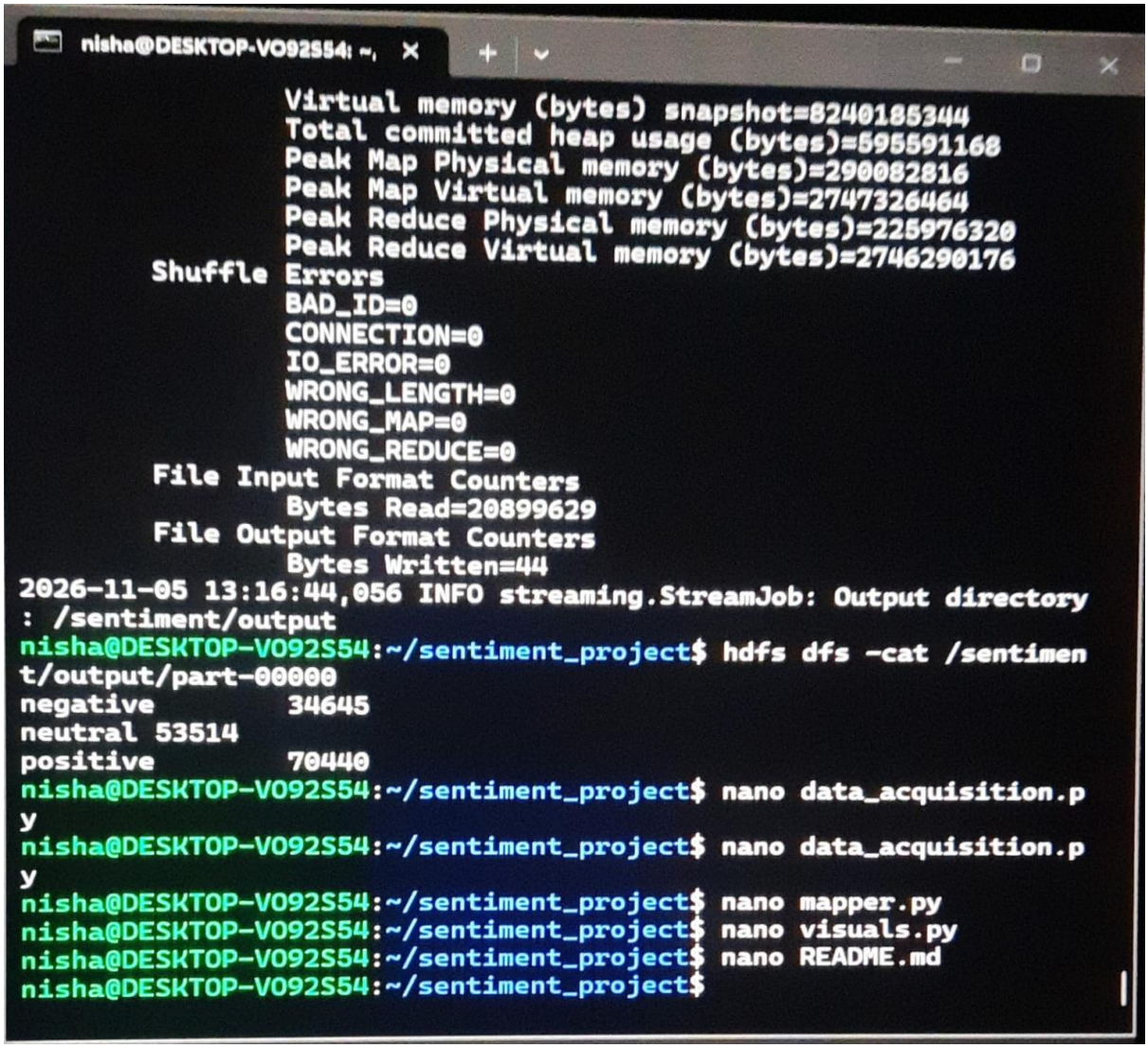
Task 5: (Data Visualization) [jupyter notebook and Hadoop as well]

Task 4: (Scalability and Distributed Processing)


nisha@DESKTOP-VO92S54: ~
nisha@DESKTOP-VO92S54:~/sentiment_project$ chmod +x mapper.py reducer.py
nisha@DESKTOP-VO92S54:~/sentiment_project$ head -n 50 Twitter_Data.csv | ./mapper.py | sort -k1,1 | ./reducer.py
negative 10
neutral 16
positive 16
nisha@DESKTOP-VO92S54:~/sentiment_project$ find ~/cse/hadoop/ -name "hadoop-streaming-*.jar"
/home/nisha/cse/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar
/home/nisha/cse/hadoop/share/hadoop/tools/sources/hadoop-streaming-3.3.6-sources.jar
/home/nisha/cse/hadoop/share/hadoop/tools/sources/hadoop-streaming-3.3.6-test-sources.jar
nisha@DESKTOP-VO92S54:~/sentiment_project$ hdfs dfs -rm -r /sentiment/output 2>/dev/null || true
nisha@DESKTOP-VO92S54:~/sentiment_project$ hadoop jar /home/nisha/cse/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar \
-files mapper.py,reducer.py \
-mapper ./mapper.py \
-reducer ./reducer.py \
-input /sentiment/input/Twitter_Data.csv \
-output /sentiment/output
packageJobJar: [/tmp/hadoop-unjar1486177636515904O826/] [] /tmp/streamjob3160047656804459120.jar tmpDir=null
2026-11-05 13:09:09,742 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-11-05 13:09:10,192 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-11-05 13:09:11,021 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/nisha/.stagi



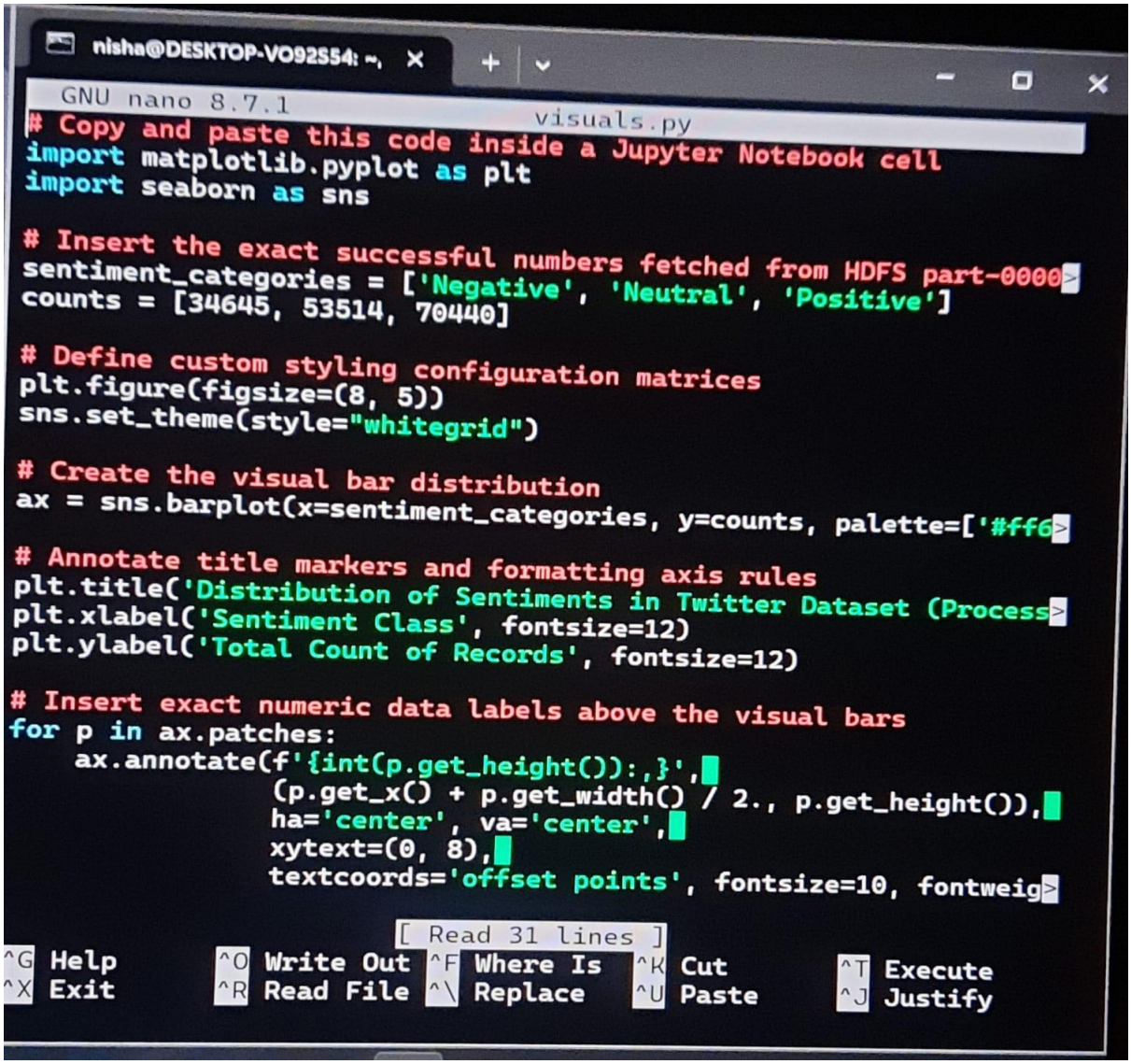
Complete Metics Extraction and view

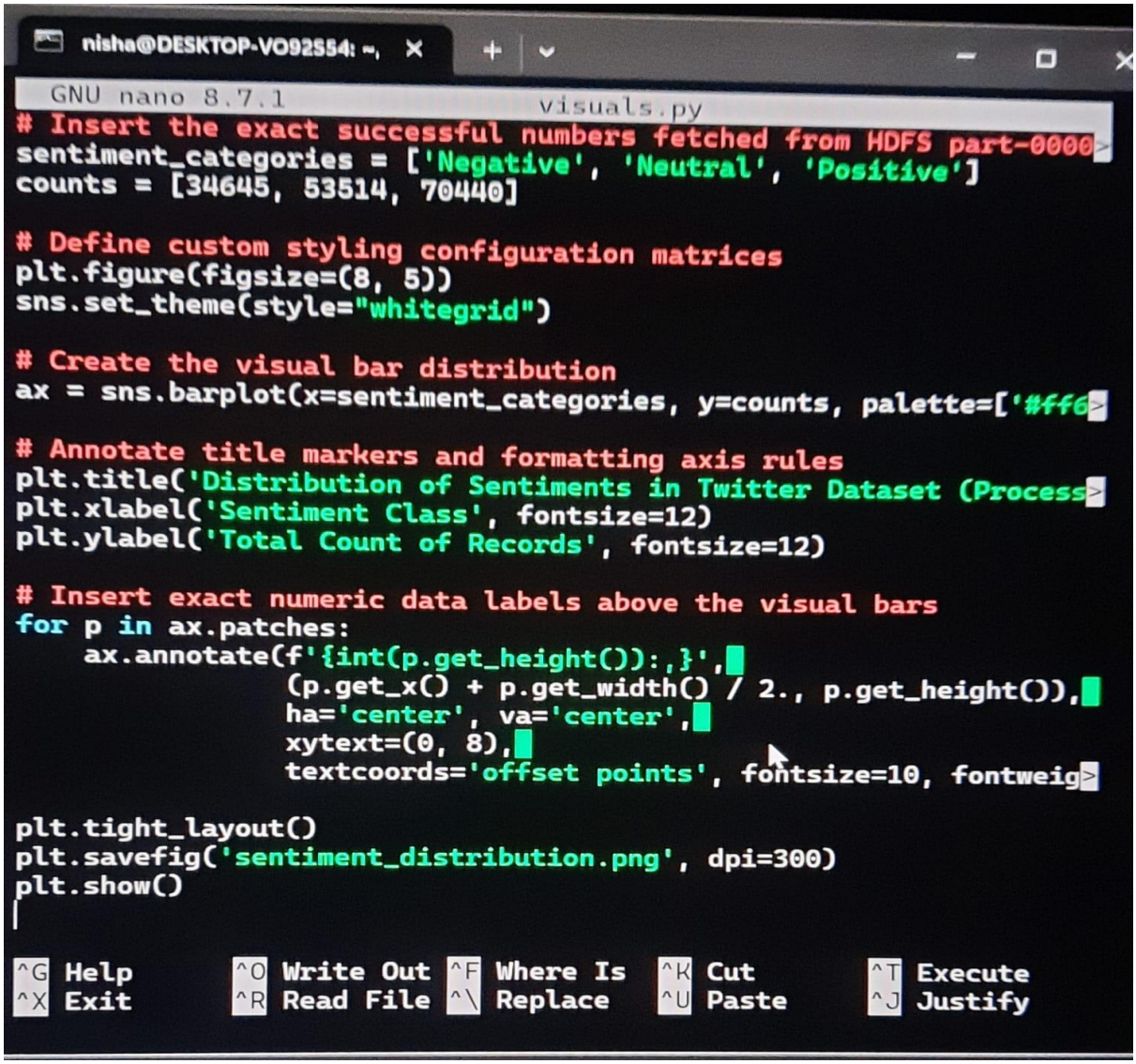


Successful Hadoop streaming!!

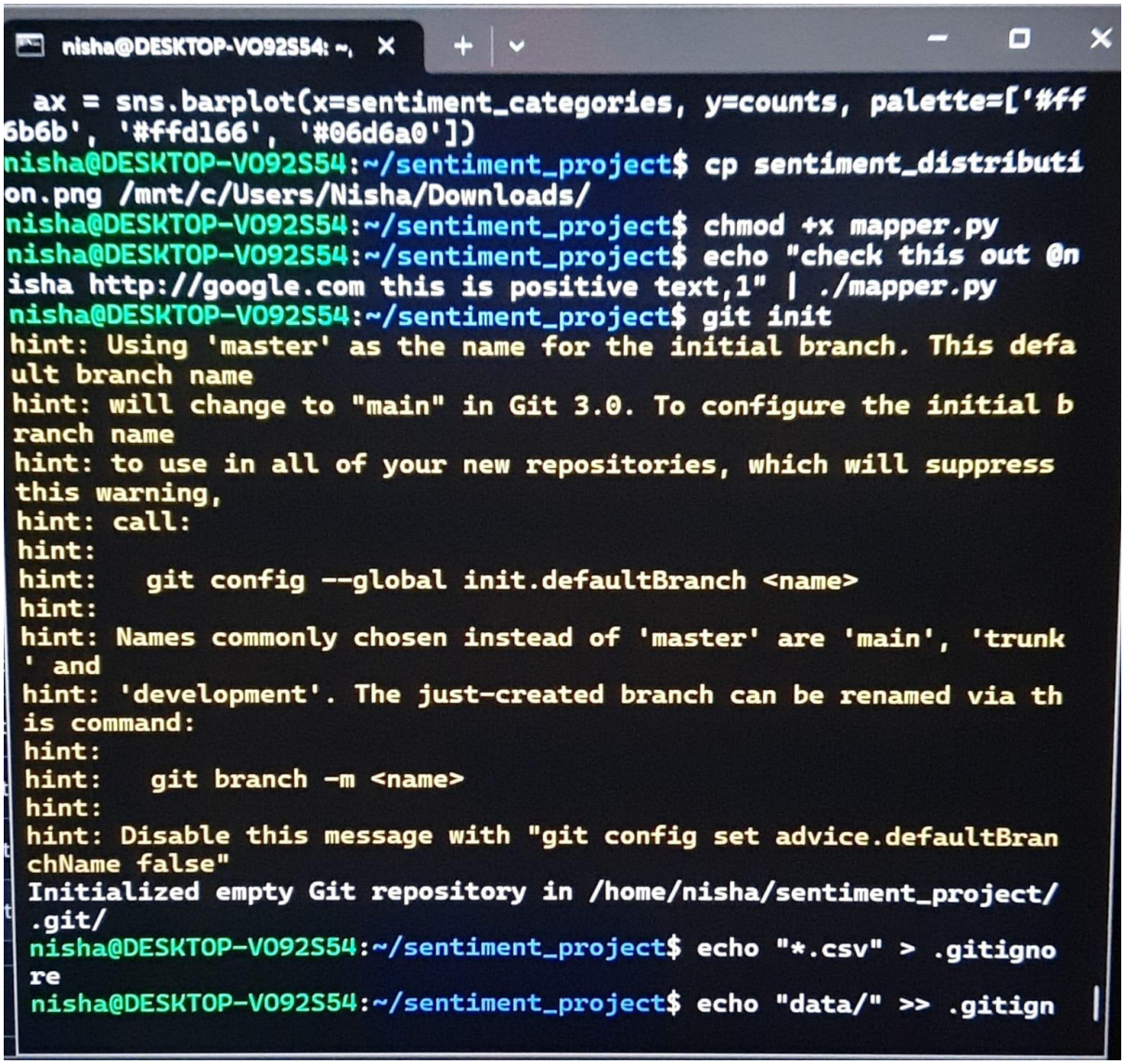
Make a directory visuals.py

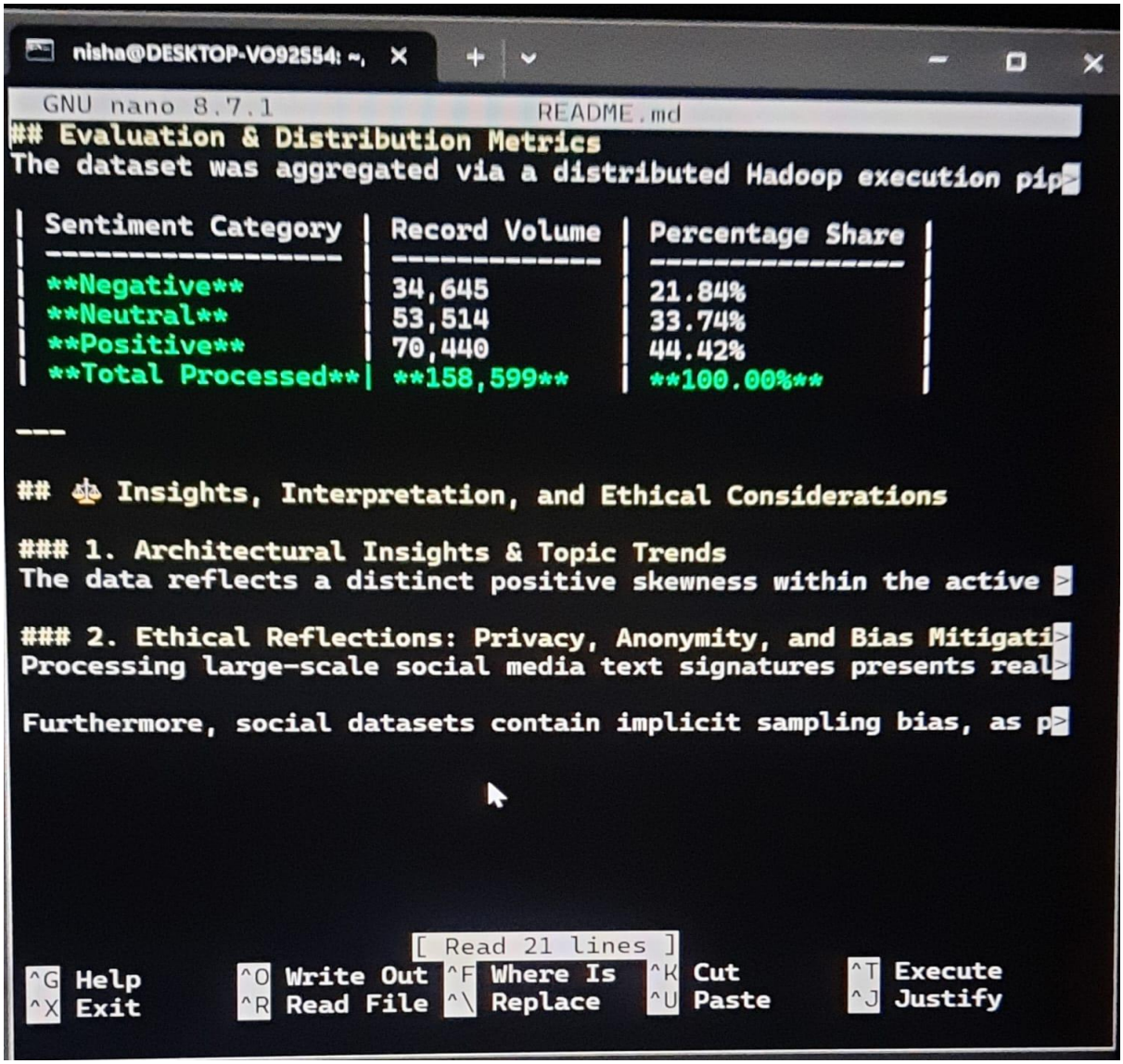
Nano visuals.py


nisha@DESKTOP-VO92554: ~
GNU nano 8.7.1
visuals.py
Copy and paste this code inside a Jupyter Notebook cell
import matplotlib.pyplot as plt
import seaborn as sns
Insert the exact successful numbers fetched from HDFS part-0000
sentiment_categories = ['Negative', 'Neutral', 'Positive']
counts = [34645, 53514, 70440]
Define custom styling configuration matrices
plt.figure(figsize=(8, 5))
sns.set_theme(style="whitegrid")
Create the visual bar distribution
ax = sns.barplot(x=sentiment_categories, y=counts, palette=['#ff6
Annotate title markers and formatting axis rules
plt.title('Distribution of Sentiments in Twitter Dataset (Process
plt.xlabel('Sentiment Class', fontsize=12)
plt.ylabel('Total Count of Records', fontsize=12)
Insert exact numeric data labels above the visual bars
for p in ax.patches:
ax.annotate(f'{int(p.get_height()):,}',
(p.get_x() + p.get_width() / 2., p.get_height()),
ha='center', va='center',
xytext=(0, 8),
textcoords='offset points', fontsize=10, fontweig
[Read 31 lines]
^G Help
^O Write Out
^F Where Is
^K Cut
^T Execute
^X Exit
^R Read File
^\ Replace
^U Paste
^J Justify

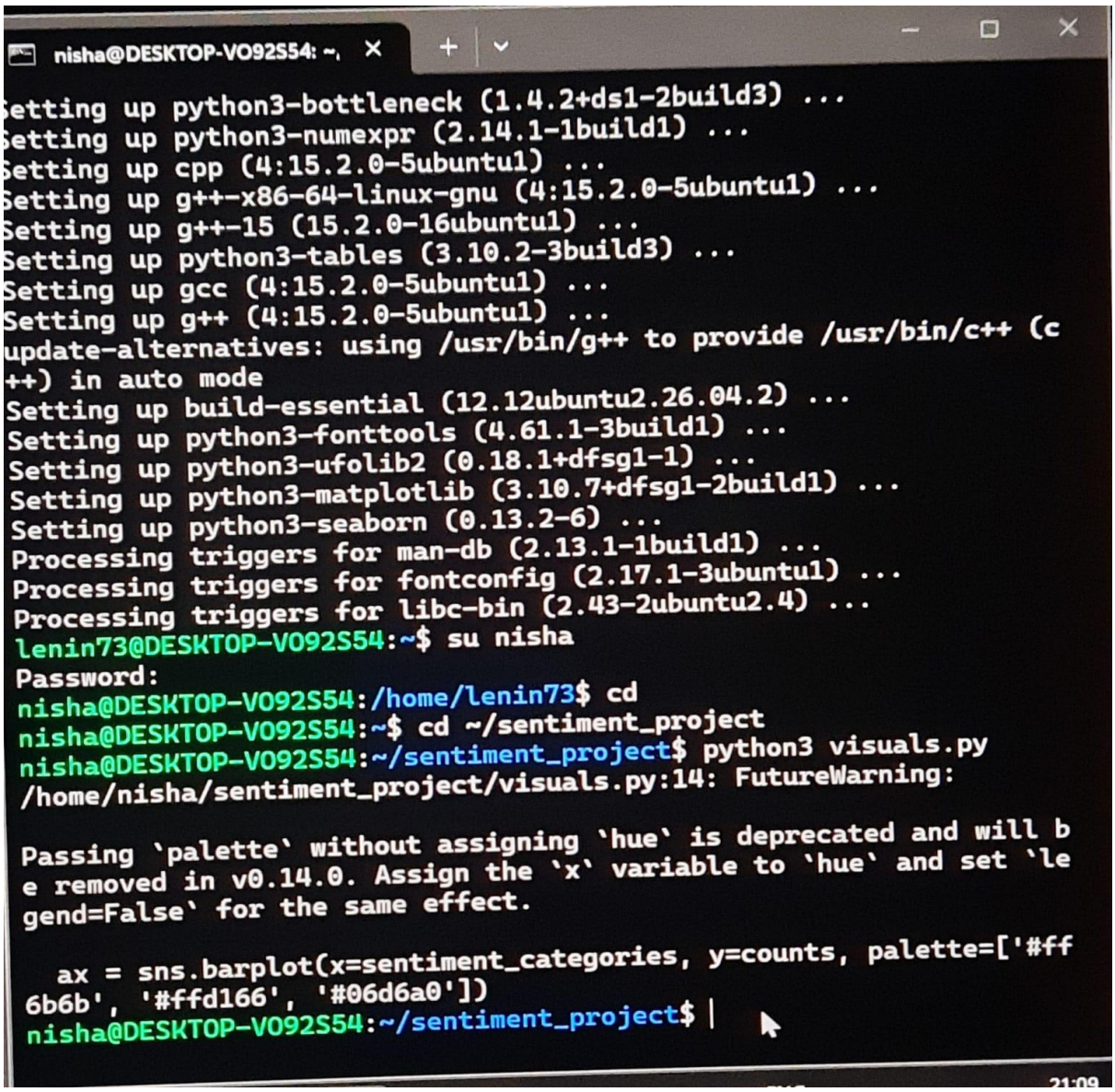


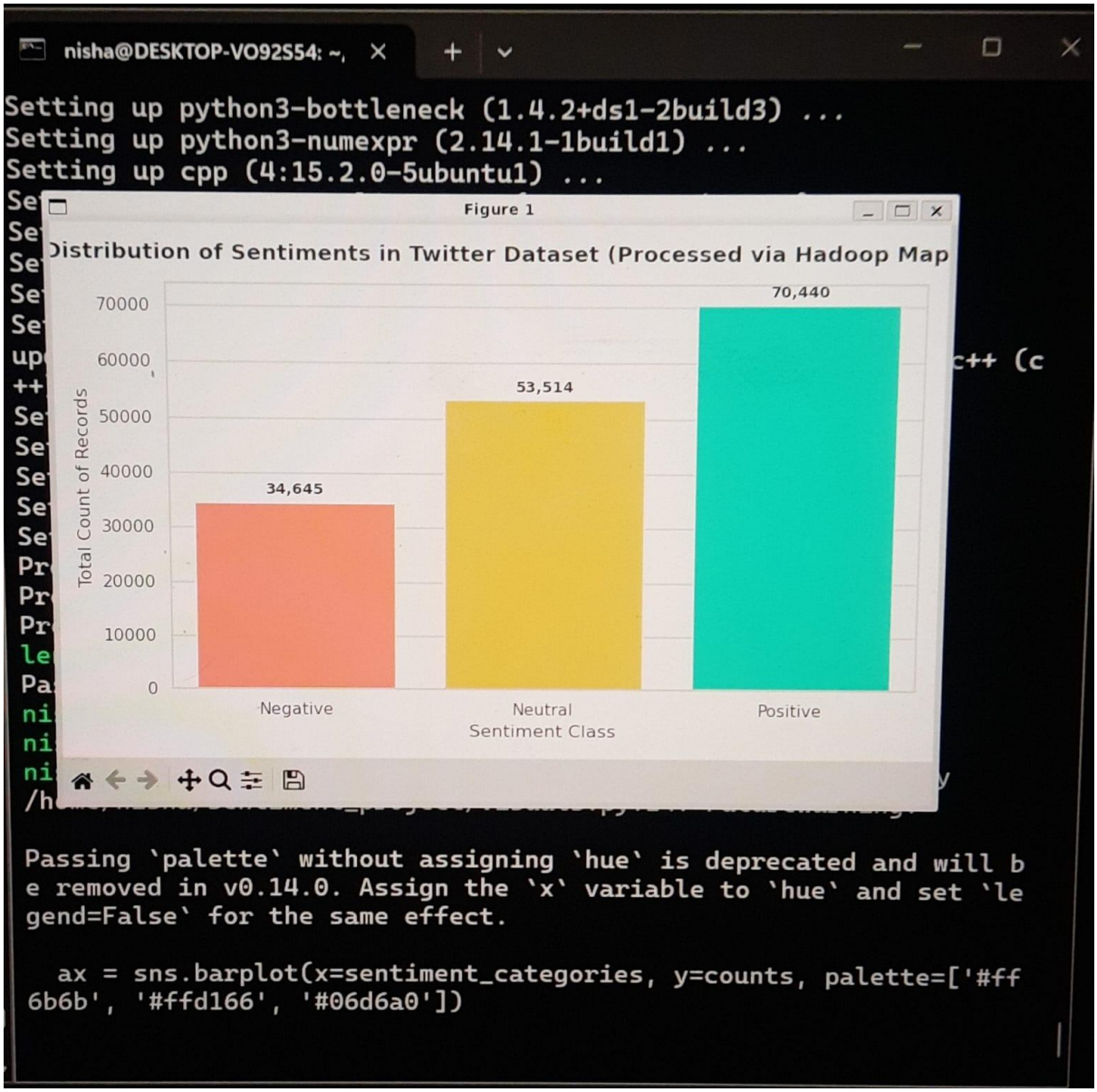
Live view through the linux terminal


nisha@DESKTOP-VO92S54: ~
ax = sns.barplot(x=sentiment_categories, y=counts, palette=['#ff6b6b', '#ffd166', '#06d6a0'])
nisha@DESKTOP-VO92S54:~/sentiment_project$ cp sentiment_distribution.png /mnt/c/Users/Nisha/Downloads/
nisha@DESKTOP-VO92S54:~/sentiment_project$ chmod +x mapper.py
nisha@DESKTOP-VO92S54:~/sentiment_project$ echo "check this out @nisha http://google.com this is positive text,1" | ./mapper.py
nisha@DESKTOP-VO92S54:~/sentiment_project$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint: git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint: git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/nisha/sentiment_project/.git/
nisha@DESKTOP-VO92S54:~/sentiment_project$ echo "*.csv" > .gitignore
nisha@DESKTOP-VO92S54:~/sentiment_project$ echo "data/" >> .gitign

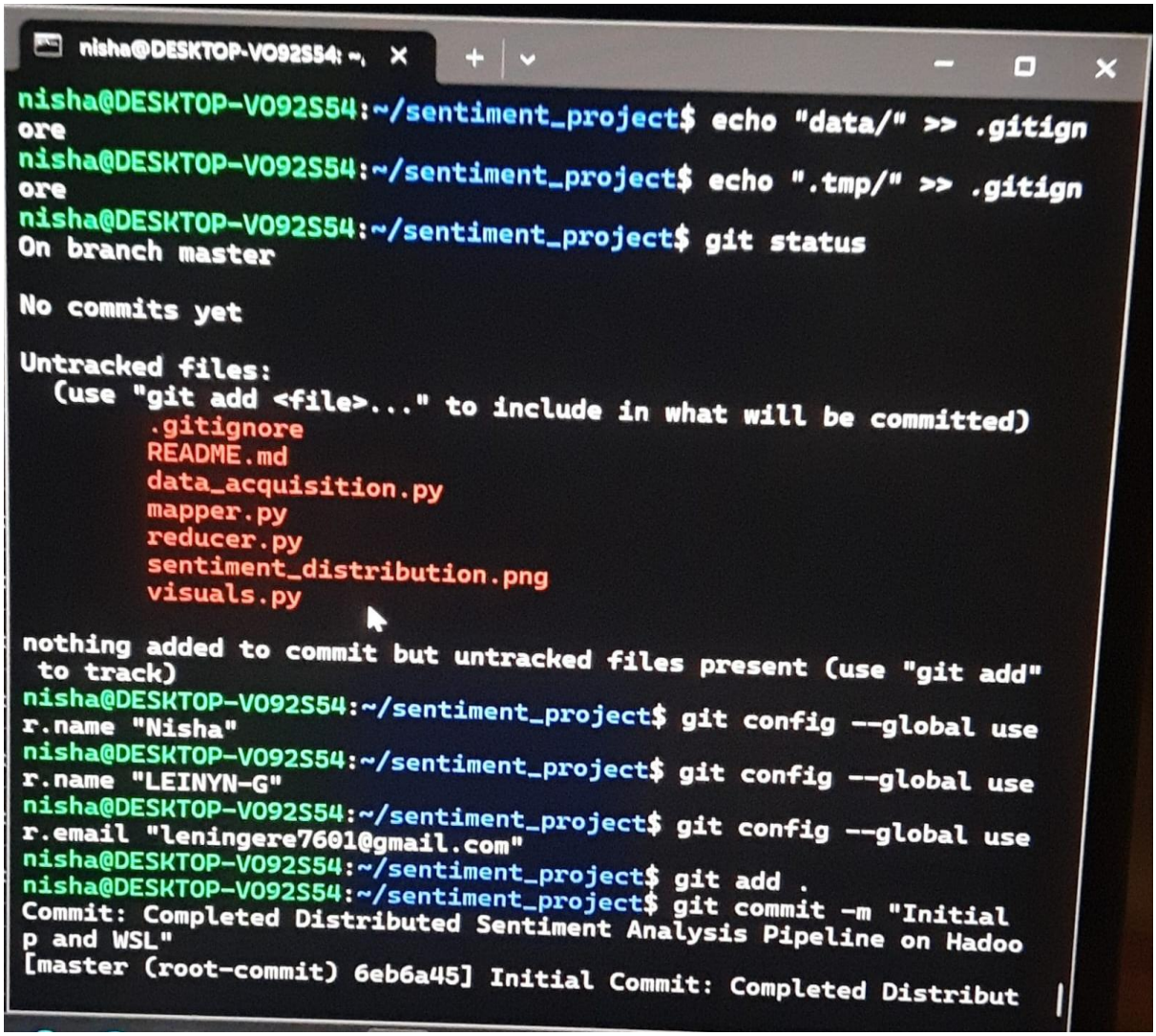

nisha@DESKTOP-VO92554: ~
GNU nano 8.7.1 README.md
Evaluation & Distribution Metrics
The dataset was aggregated via a distributed Hadoop execution pip
Sentiment Category	Record Volume	Percentage Share
Negative	34,645	21.84%
Neutral	53,514	33.74%
Positive	70,440	44.42%
Total Processed	**158,599**	**100.00%**

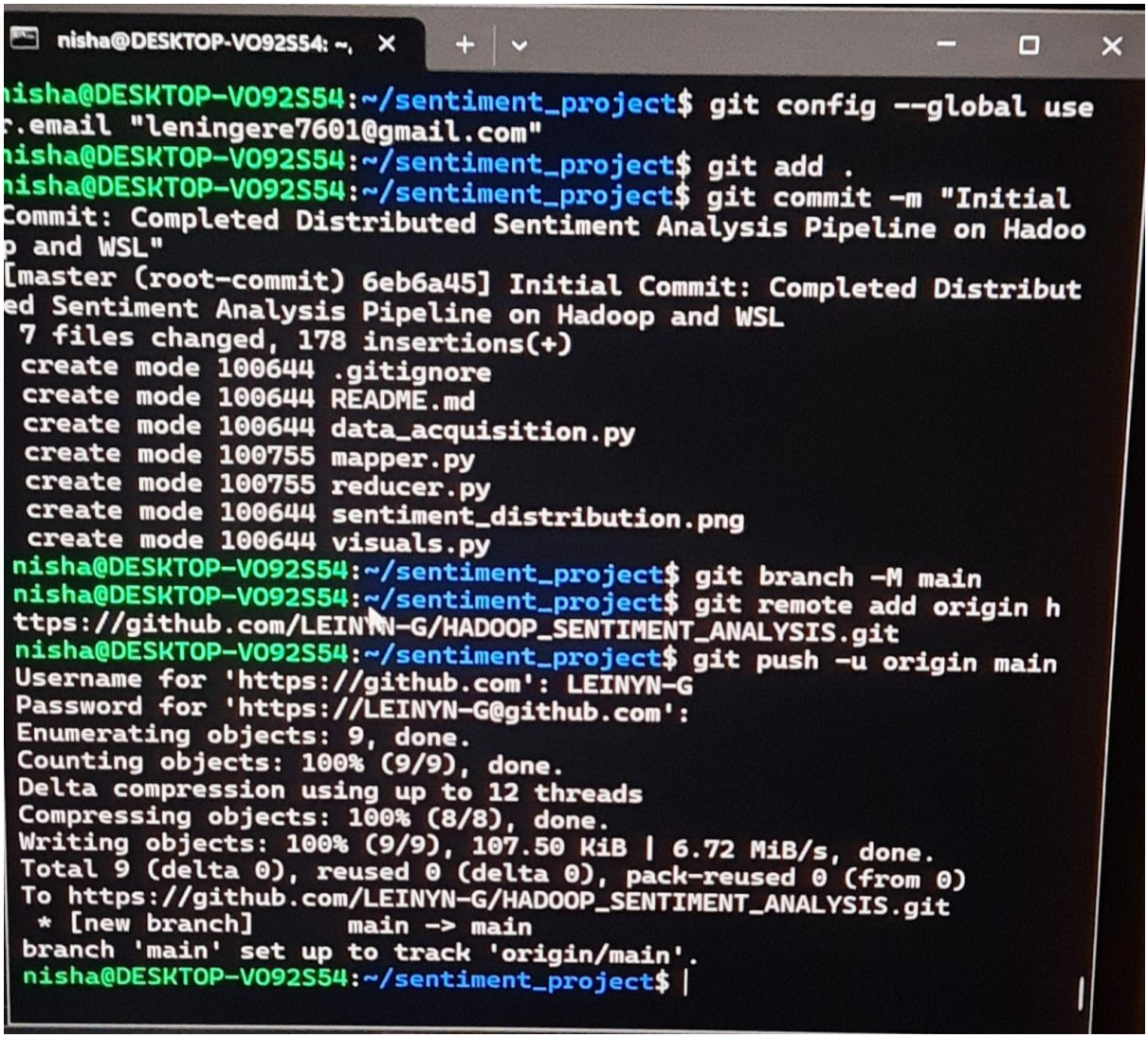
⚖️ Insights, Interpretation, and Ethical Considerations
1. Architectural Insights & Topic Trends
The data reflects a distinct positive skewness within the active
2. Ethical Reflections: Privacy, Anonymity, and Bias Mitigati
Processing large-scale social media text signatures presents real
Furthermore, social datasets contain implicit sampling bias, as p
[Read 21 lines]
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify


nisha@DESKTOP-VO92S54: ~
Setting up python3-bottleneck (1.4.2+ds1-2build3) ...
Setting up python3-numexpr (2.14.1-1build1) ...
Setting up cpp (4:15.2.0-5ubuntu1) ...
Setting up g++-x86-64-linux-gnu (4:15.2.0-5ubuntu1) ...
Setting up g++-15 (15.2.0-16ubuntu1) ...
Setting up python3-tables (3.10.2-3build3) ...
Setting up gcc (4:15.2.0-5ubuntu1) ...
Setting up g++ (4:15.2.0-5ubuntu1) ...
update-alternatives: using /usr/bin/g++ to provide /usr/bin/c++ (c++) in auto mode
Setting up build-essential (12.12ubuntu2.26.04.2) ...
Setting up python3-fonttools (4.61.1-3build1) ...
Setting up python3-ufolib2 (0.18.1+dfsg1-1) ...
Setting up python3-matplotlib (3.10.7+dfsg1-2build1) ...
Setting up python3-seaborn (0.13.2-6) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for fontconfig (2.17.1-3ubuntu1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.4) ...
lenin73@DESKTOP-VO92S54:~$ su nisha
Password:
nisha@DESKTOP-VO92S54:/home/lenin73$ cd
nisha@DESKTOP-VO92S54:~$ cd ~/sentiment_project
nisha@DESKTOP-VO92S54:~/sentiment_project$ python3 visuals.py
/home/nisha/sentiment_project/visuals.py:14: FutureWarning:
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.
ax = sns.barplot(x=sentiment_categories, y=counts, palette=['#ff6b6b', '#ffd166', '#06d6a0'])
nisha@DESKTOP-VO92S54:~/sentiment_project$



Push it all in Git:


nisha@DESKTOP-VO92S54:~/sentiment_project$ echo "data/" >> .gitign
ore
nisha@DESKTOP-VO92S54:~/sentiment_project$ echo ".tmp/" >> .gitign
ore
nisha@DESKTOP-VO92S54:~/sentiment_project$ git status
On branch master
No commits yet
Untracked files:
(use "git add <file>..." to include in what will be committed)
.gitignore
README.md
data_acquisition.py
mapper.py
reducer.py
sentiment_distribution.png
visuals.py
nothing added to commit but untracked files present (use "git add"
to track)
nisha@DESKTOP-VO92S54:~/sentiment_project$ git config --global use
r.name "Nisha"
nisha@DESKTOP-VO92S54:~/sentiment_project$ git config --global use
r.name "LEINYN-G"
nisha@DESKTOP-VO92S54:~/sentiment_project$ git config --global use
r.email "leningere7601@gmail.com"
nisha@DESKTOP-VO92S54:~/sentiment_project$ git add .
nisha@DESKTOP-VO92S54:~/sentiment_project$ git commit -m "Initial
Commit: Completed Distributed Sentiment Analysis Pipeline on Hadoo
p and WSL"
[master (root-commit) 6eb6a45] Initial Commit: Completed Distribut


nisha@DESKTOP-VO92S54: ~,
nisha@DESKTOP-VO92S54:~/sentiment_project$ git config --global use
r.email "leningere7601@gmail.com"
nisha@DESKTOP-VO92S54:~/sentiment_project$ git add .
nisha@DESKTOP-VO92S54:~/sentiment_project$ git commit -m "Initial
Commit: Completed Distributed Sentiment Analysis Pipeline on Hadoo
p and WSL"
[master (root-commit) 6eb6a45] Initial Commit: Completed Distribut
ed Sentiment Analysis Pipeline on Hadoop and WSL
7 files changed, 178 insertions(+)
create mode 100644 .gitignore
create mode 100644 README.md
create mode 100644 data_acquisition.py
create mode 100755 mapper.py
create mode 100755 reducer.py
create mode 100644 sentiment_distribution.png
create mode 100644 visuals.py
nisha@DESKTOP-VO92S54:~/sentiment_project$ git branch -M main
nisha@DESKTOP-VO92S54:~/sentiment_project$ git remote add origin h
ttps://github.com/LEINYN-G/HADOOP_SENTIMENT_ANALYSIS.git
nisha@DESKTOP-VO92S54:~/sentiment_project$ git push -u origin main
Username for 'https://github.com': LEINYN-G
Password for 'https://LEINYN-G@github.com':
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 12 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (9/9), 107.50 KiB | 6.72 MiB/s, done.
Total 9 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/LEINYN-G/HADOOP_SENTIMENT_ANALYSIS.git
* [new branch] main -> main
branch 'main' set up to track 'origin/main'.
nisha@DESKTOP-VO92S54:~/sentiment_project$